

CIRCULAR DECISION ENGINE

RePurpose AI

Make circular economy decisions accessible to everyone.

FROM UNWANTED ITEM

→ safer, higher-value next action

SDG 12 · everyday decision support

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THE PROBLEM

Most unwanted items still have a better next step

MUNICIPAL SOLID WASTE

2.01

billion tonnes

generated annually

01 · ITEM

Plain-language description

People discard reusable, repairable, or donatable items because the right action is unclear.

02 · ACTION

Actionable recommendation

RePurpose AI turns a plain-language item description into an actionable, cited recommendation.

03 · PATH

Next action

Reuse · repair · donate · recycle · responsible disposal.

ONE QUESTION, MANY CONTEXTS

One decision engine can serve every kind of unwanted item

What should I do with this?

Households

Everyday unwanted items

College Move-Outs

Furniture and small goods

Small Businesses

Surplus inventory
decisions

NGO Triage

Sorting donated goods

How might we help people make informed circular economy decisions so fewer items end up in landfill?

Meet users at the moment they ask, "What should I do with this?"

Made with GAMMA

A lightweight stack keeps the prototype fast to extend

01 INTERFACE

Streamlit Python web app

02 REASONING

Groq API with compound-mini

plus a custom RAG and conversational AI pipeline.

03 RETRIEVAL

sentence-transformers all-MiniLM-L6-v2

ChromaDB as a local persistent vector store.

04 KNOWLEDGE

PyPDF2 and plain text

configured with python-dotenv. No LangChain dependency.

Every recommendation follows the same evidence path

01

User describes the item

02 · HAZARD GATE

Keyword check

Batteries · chemicals · electronics · other risks

03

Embed + match

ChromaDB knowledge base

04

Retrieve context

Groq language model

05 · OUTPUT

Pathway + reasoning

Sources · next steps · confidence

Knowledge areas: circular economy frameworks · e-waste · furniture · textiles · plastics · hazardous disposal.

AI chooses the highest-value pathway that still fits the item

01 Reuse as-is

Functional and ready for another user.

02 Repair

Fixable damage restores value.

03 Repurpose / upcycle

The old form gets a new function.

04 Donate

Functional but unwanted items move to someone who needs them.

05 Refurbish

Professional restoration prepares the item for resale.

06 Recycle

Materials are recoverable, but the item is not reusable.

07 Responsible disposal

The last resort, through certified channels only.

RePurpose AI turns SDG 12 into a practical everyday decision

Primary: SDG 12, Responsible Consumption and Production.

SDG 12.5: reduce waste through prevention, reduction, recycling, and reuse.

SDG 12.8: give people information for sustainable lifestyles.

Secondary links: SDG 13 Climate Action, SDG 11 Sustainable Cities, and SDG 9 Industry and Innovation.

The prototype is designed to be useful, grounded, and extensible

Control	What it does	Evidence
Grounded	Every recommendation cites sustainability documents.	Curated knowledge base
Hazard-aware	Safety warnings appear before ordinary reuse advice.	Hazard keyword check
Hierarchy-enforced	The model is instructed to prefer higher-value circular options.	Pathway hierarchy
Multi-turn	Users can ask follow-up questions in the same session.	Conversational AI pipeline
Lightweight	The app runs locally without cloud infrastructure.	Streamlit Python web app
Extensible	Adding documents to the knowledge base expands the system after re-ingestion.	Six manually curated documents

NEXT STEP

The next step is moving from text-only guidance to local action

TODAY

Text-only guidance

Static knowledge base · text-only input · no location awareness.

Dependency on API availability and free-tier rate limits.

NEXT

Local action

Image upload · location-aware recommendations · multilingual support.

Impact dashboard · scheduled knowledge-base updates.

BUILD TIMELINE

Two AI tools accelerated the build; curated knowledge kept it accountable

01 PROBLEM FRAMING

IBM Bob

Ideation · How Might We · SDG alignment · solution architecture

02 IMPLEMENTATION

Kiro

Architecture · coding · debugging · end-to-end deployment

03 KNOWLEDGE BASE

Grounded prototype

Six manually curated documents covering circular economy, e-waste, furniture, textiles, plastics, and hazardous disposal.

THE ASK

Make circular economy decisions accessible to everyone

RePurpose AI helps people move from “throw it away” to a safer, higher-value next action.

GitHub: github.com/varshapadmanaban05/AI-Internship-on-sustainability

Built with IBM Bob and Kiro · 1M1B · IBM SkillsBuild internship · SDG 12.